The Swap Rule in Crystals

Rock salt, cadmium iodide, zinc blende and the Laves phases on every stacking

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The ideas and names are DICTO's. Computation and proof with Claude (Anthropic).

Abstract. Many crystals come in a cubic and a hexagonal form that differ only in how their layers are stacked, and in many mixed stackings between them. Paper #57 found that at close packing the space of every stacking is shared out into pieces borrowed from the two pure stackings, whole or in halves (the rule of halves), and #61 proved the Swap Rule: the two mixed pieces always add up to the two pure ones. Here we take both rules to four classical crystal families and prove, exactly, that **the Swap Rule holds in all four**, in all 299 cases where it can be tested: rock salt and nickel arsenide (NaCl, NiAs), the half-filled layers of cadmium chloride and cadmium iodide ($CdCl_2$, CdI_2), zinc blende and wurtzite with two different atoms (ZnS), and the Laves phases ($MgCu_2$, $MgZn_2$, $MgNi_2$). The rule of halves holds in full for rock salt and cadmium iodide: each octahedral piece is exactly half cubic and half hexagonal, each ball and tetrahedral piece is borrowed whole from the layer it rests on. In zinc blende the atoms keep the rule of halves, while the empty holes stop being fixed mixtures and still swap exactly. In the Laves phases, near the real alloys, every stacking is cut into exactly the same pieces.

1. The idea

DICTO's rule of halves (#57) and Swap Rule (#61) were found on the plain close-packed stacking of equal balls. DICTO asked whether they belong to that one structure or to stacking itself. The test is to fill the holes of the stacking with atoms as real crystals do, and to ask the same two questions of every resulting piece: is it borrowed from the pure cubic and hexagonal stackings, and do the mixed pieces always add up to the pure ones?

2. Set-up

A close-packed stacking is a word in the layer positions A, B, C; each layer is cubic (c) when its two neighbours differ and hexagonal (h) when they are the same. Every gap between two layers X and Y has a ball site at its foot, two tetrahedral sites (one over Y at ¼ of the gap, one over X at ¾) and one octahedral site (over the third position at ½). A crystal family fills some of these sites with atoms. Its space is shared out by weight (a power diagram, as in #57): atoms of one kind have weight λ, empty holes weight 0, and λ runs over the whole range in which every site keeps a piece. The four families:

Family	Atoms (weight λ)	Empty (weight 0)	Pure cubic / hexagonal
Rock salt / NiAs	balls, octahedral sites	tetrahedral sites	NaCl / NiAs
$CdCl_2$ / CdI_2	balls, octahedral sites in every other gap	tetrahedral sites, the other octahedral sites	$CdCl_2$ / CdI_2
Zinc blende / wurtzite	balls (λ), one tetrahedral site per gap (λ + μ, a second atom; μ = ±1/20, ±1/10)	octahedral and the other tetrahedral sites	ZnS sphalerite / wurtzite
Laves phases	big atoms on the diamond-type sites (λ)	— (small atoms, weight 0: a kagome of three and one more per gap)	$MgCu_2$ (C15) / $MgZn_2$ (C14); $MgNi_2$ (C36) mixed

The Laves network is built on every stacking by one rule (§5) and checked exactly: every small atom has 6 small and 6 big neighbours (an icosahedron), every big atom 4 big and 12 small, at squared distances ¼, 11/32 and 3/8 (ball spacing 1), the ideal Laves geometry. On the cubic stacking it is exactly $MgCu_2$.

3. Main theorem

> **Theorem.** In each of the four families, on every close-packed stacking of up to six layers (all of them, up to symmetry; odd ones doubled for $CdCl_2$/CdI_2) and at every weight λ of its range, every piece has its volume fixed by the layers round its gap (four layers; six for the Laves foot atom), and
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> (1) **Swap Rule.** For every kind of piece whose volume is fixed by the types of its two neighbouring layers, V(c under h) + V(h under c) = V(c under c) + V(h under h): 299 of 299 cases, none failing. (Not testable: 40 zinc blende cases where a piece vanishes in one pure stacking, and the Laves foot atom above λ = 17/96, which reads three layer types.)
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> (2) **Rule of halves, rock salt and cadmium iodide.** Every octahedral piece, filled or empty, is exactly half the cubic piece and half the hexagonal piece. Every ball takes the pure piece of its own layer's type, and every tetrahedral piece that of the layer holding its base.
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> (3) **Zinc blende.** The balls and the second atoms keep the rule of halves (the balls in 40 and the second atoms in 34 of the 43 windows where both pure pieces exist; the balls are undetermined in the other 3), the empty octahedral holes are a fixed cubic–hexagonal mixture in only 4 of the 43, yet every piece swaps exactly.
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> (4) **Laves phases.** For $-11/32 < \lambda < 17/96$, which contains the plain Voronoi cells of the real alloys (λ = 0), every Laves stacking is cut into exactly the same pieces: the cubic and hexagonal pieces are equal. Above 17/96 five of the six kinds of piece keep the rule of halves and swap exactly; the big atom at the foot of each gap reads three layer types, not two.

4. The results family by family

Family	Weight range	Windows	Swap Rule	Rule of halves
Rock salt / NiAs	−3/8 to 3/8	5	20 of 20	in full: octahedral ½c + ½h, balls and tetrahedral whole
$CdCl_2$ / CdI_2	−3/8 to 3/8	10	80 of 80	in full, filled and empty octahedral sites alike
Zinc blende / wurtzite, four values of μ	within −3/8 to 3/8	53	172 of 172	atoms yes; empty holes mostly not fixed mixtures
Laves phases	−11/32 to 11/32	5	27 of 27	identical pieces below 17/96; above, 5 of 6 kinds

The rock salt and cadmium iodide pattern is exactly #57's: three balls from each side make an octahedral piece half of each, a tetrahedral piece with its base in one layer belongs to that layer. Zinc blende repeats what #61 found below half the squash: the Swap Rule outlasts the rule of halves, because each piece's volume stays a sum of one share from each neighbouring layer even when the shape is no longer a fixed mixture.

5. The Laves network on every stacking

In each gap X, Y (third position Z) the big atoms sit over X at 0 and ¾ of the gap, as the diamond-type atoms of #60; the small atoms form a kagome over X shifted by half a ball spacing, (½, 0), (0, ½), (½, ½), at ⅜, and one more over Z at ⅞. On the cubic stacking this rule gives the same small atoms as the classical construction of $MgCu_2$ (each empty hole of the big-atom diamond surrounded by an inverted tetrahedron of small atoms), checked exactly; on the hexagonal stacking it gives $MgZn_2$, and on every stacking up to six layers it gives the ideal Frank–Kasper neighbourhoods, with the next neighbours clearly farther (squared gap 31/96 or more).

6. Proof

For each family and each stacking word we run the exact weight walk of #57 (rational arithmetic throughout): it finds every λ at which some piece changes its combinatorial type and, between these stops, every piece's volume as an exact polynomial in λ, with the engine's own checks (pieces tile the cell, faces matched, distance bounds) — no check failed. Locality: two stackings that agree on the four layers round a gap (six for the Laves foot atom) give the same volume, every time (0 clashes). Each piece's volume is then compared, as a polynomial, with a·V(cubic) + b·V(hexagonal) for constant rational a, b (the rule of halves), and the swap sum is checked as a polynomial identity. Scripts and logs: study 2026-10-27-splash-rules, run2/ (rocksalt_mix.py, cdi2_mix.py, zincblende_mix.py, laves_net.py, laves_mix.py, laves_mix6.py).

7. Prior art and credit

Polytypes are classical: Ramsdell's notation (3C, 2H, 4H …) and the c/h notation of Jagodzinski; the cubic and hexagonal forms of ZnS, SiC, CdI_2 and the NaCl–NiAs pair (Verma and Krishna, *Polymorphism and Polytypism in Crystals*, 1966). The Laves phases C15, C14 and C36 are polytypes of one common slab, with c, h and ch stackings, and are built from Z12 and Z16 Frank–Kasper clusters. A study of block-polymer Laves phases found the free energy of C36 to be almost exactly the average of C14 and C15, a statement of the same shape as the

Swap Rule, but for energies and not proved exactly. The weight walk, the rule of halves and the Swap Rule are #57 and #61. We found no earlier statement that the pieces of these crystals obey the rule of halves or the Swap Rule exactly.

> **Sources checked** (28 October 2026). Reference library (ask-library on "Laves phase polytypes C14 C15 C36 stacking; polytypism CdI_2 SiC ZnS Verma Krishna; NiAs rock salt stacking; Voronoi or power diagram of polytypes; Frank–Kasper polyhedra Z12 Z16"): J. Yi and G. S. Canright, "Possible disordered ground states for close-packed polytypes" (1995); E. Abe et al., long-period stacking polytypes in Mg–Zn–Y (2011); A. Haji-Akbari et al., densely packed tetrahedra (2009) and phase diagram of hard tetrahedra (2011); Cadotte et al., hard tetrahedra and octahedra (2016); T. Fujita et al., cluster packing of F-type icosahedral (2013). Web search "Laves phase polytypes C14 C15 C36 stacking sequence Voronoi cell volume Frank–Kasper polyhedra": arXiv 2103.09514 (Laves phase crystal analysis, LaCA); par.nsf.gov 10299033 and 10321376 and RSC Soft Matter d1sm01063h (Laves phases in block polymers, C36 free energy as the average of C14 and C15); APS March Meeting 2022 K16.8 (absence of C36 in diblock melts); open.fau.de 22393; MPI CPfS Laves phase project page; poivs.tsput.ru conference talk. The Verma–Krishna book and Ramsdell's and Jagodzinski's notations are cited from standard knowledge; not read for this paper.

8. Conclusion

The Swap Rule is not a property of one packing. In rock salt, cadmium iodide, zinc blende and the Laves phases, on every stacking, the two mixed pieces add up to the two pure ones, exactly, in every one of the 299 cases where both pure pieces exist and two layer types fix the piece. Where atoms sit in octahedral sites the pieces are DICTO's Janus halves, as at close packing; where the holes stay empty the shapes may wander but the volumes still swap; and in the Laves phases the stacking, near the real alloys, makes no difference to the pieces at all. One rule runs through all four families.

> **Data and code.** Study 2026-10-27-splash-rules (run2/), #57's exact weight-walk engine unchanged. References: DICTO, The Rule of Halves (#57, DOI 10.5281/zenodo.23647898); DICTO Janus c/h Pyrochlore (#60, 10.5281/zenodo.23678750); The Swap Rule (#61, 10.5281/zenodo.23682596).